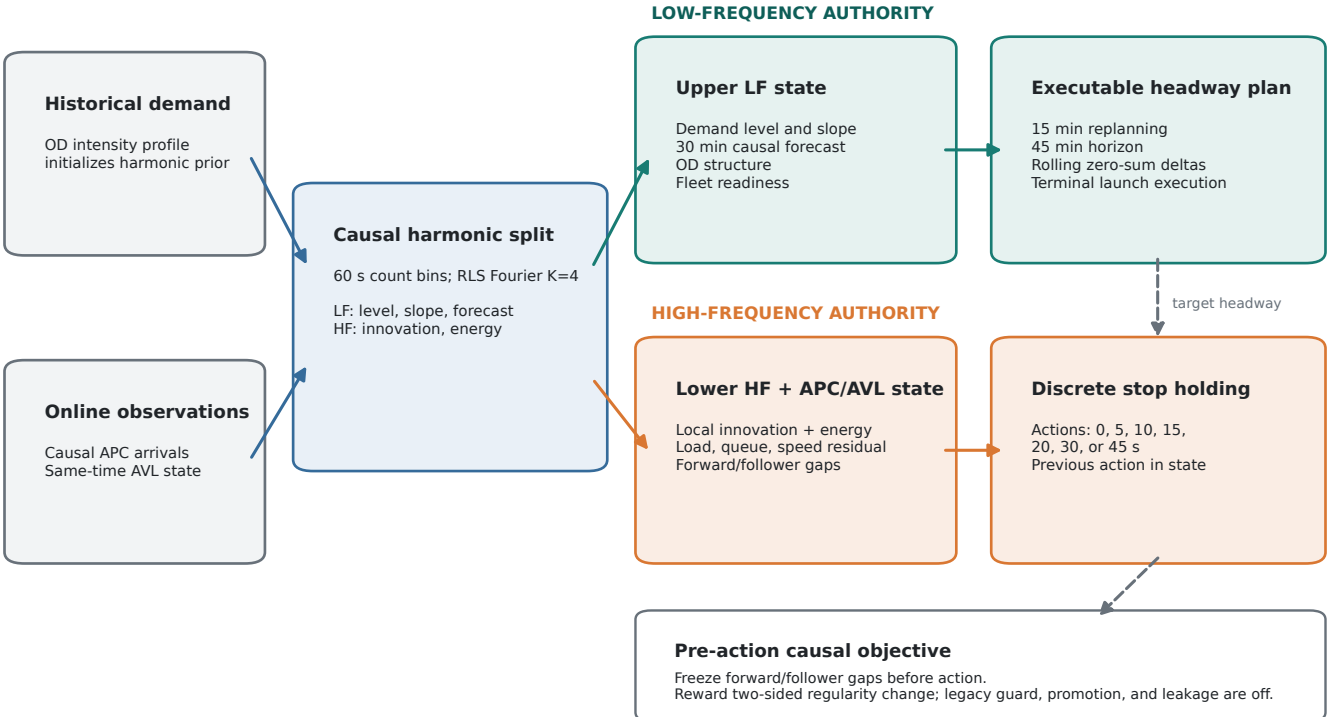
LOW-FREQUENCY AUTHORITY
Historical demand
OD intensity profile
initializes harmonic prior
Online observations
Causal APC arrivals
Same-time AVL state
Causal harmonic split
60 s count bins; RLS Fourier K=4
LF: level, slope, forecast
HF: innovation, energy
Upper LF state
Demand level and slope
30 min causal forecast
OD structure
Fleet readiness
Executable headway plan
15 min replanning
45 min horizon
Rolling zero-sum deltas
Terminal launch execution
target headway
HIGH-FREQUENCY AUTHORITY
Lower HF + APC/AVL state
Local innovation + energy
Load, queue, speed residual
Forward/follower gaps
Discrete stop holding
Actions: 0, 5, 10, 15,
20, 30, or 45 s
Previous action in state
Pre-action causal objective
Freeze forward/follower gaps before action.
Reward two-sided regularity change; legacy guard, promotion, and leakage are off.